Reading the Missing Intercept

A research proposal

9 October 2026

1 The idea

The missing-intercept problem limits what cross-sectional estimates can tell us about aggregate effects (see Moll, 2025, for an exposition). For instance, suppose a programme transfers income from households with a low propensity to consume to households with a high propensity to consume. The recipients spend more of each additional euro than the households financing the programme, raising demand and generating earnings that support further consumption. A typical empirical approach would compare consumption changes across household groups while including time fixed effects to absorb aggregate movements. These fixed effects also absorb any response to the programme shared by all households, such as a common increase in consumption financed by the additional earnings it generates. This leaves part of its general-equilibrium effect unidentified even when the relative response is estimated correctly. But what if we could use the structure of an economic model to recover some of this feedback from the differences in household responses that we do observe? For instance, could information on households' spending propensities help us establish how much of their consumption response is financed by the additional earnings the programme generates?

Wolf (2023) tackles part of this problem by combining household consumption estimates with fiscal-multiplier evidence to recover aggregate responses. His demand-equivalence result establishes conditions under which private spending and government purchases generate the same equilibrium feedback. Applying it requires a fiscal comparison with matching demand paths, financing and macroeconomic conditions, alongside assumptions ensuring equivalent transmission. These requirements can be difficult to justify in a particular application, although Wolf also examines how departures from exact equivalence affect the recovery.

Manuel (2026) approaches the problem from an econometric perspective, which provides the inspiration for this proposal. He shows that households' heterogeneous responses to general-equilibrium effects can enter cross-sectional slopes, preventing them from identifying direct effects even when the aggregate shock is exogenous. His analysis makes explicit how identification depends on the comparison made across households and the controls included in the regression.

My idea is to combine these approaches by using the structure of an economic model to derive restrictions for econometric identification, for instance of the part of the consumption response financed by additional earnings. Starting from a specified model, such as a New Keynesian model with heterogeneous households, I would use household decision rules and market clearing to investigate whether differences in observed consumption and earnings responses allow us to identify how much direct spending and income feedback each contribute to the overall consumption response. For example, if we know households' spending propensities, each possible consumption response implies an income change needed to support it. Requiring these changes to be consistent with the income generated by additional spending may allow us to recover the common response without first identifying how the earnings are distributed. The aim is to establish which components

of feedback we can recover and what assumptions are needed, including when suitable fiscal evidence is unavailable.

In this proposal, I first outline the idea through a simple toy model: two household types have known marginal propensities to consume (MPCs), but receive unknown shares of the income generated by a transfer programme. Their observed spending difference alone leaves the aggregate response undetermined. However, household spending rules and market clearing supply an additional restriction: the income needed to support their consumption must equal the income generated by their purchases. Combining this restriction with the observed spending difference recovers aggregate consumption and, after subtracting the direct spending response implied by the transfers and MPCs, identifies the contribution of income feedback.

To finish this proposal, I outline a pilot regional application based on Chodorow-Reich, Nenov and Simsek (2021), where they find that additional hours generate earnings for hand-to-mouth households, while higher wages also reach stockholders who save part of the gain. In a simplified adaptation of their model, I show how differences in wage adjustment across groups can identify direct wealth spending and income feedback without supplying a local multiplier, provided household spending behaviour is common and the propensity to spend extra hours income is known. Using their replication data, I find positive payroll and employment responses to stock-wealth gains, with payroll rising by more than employment. These results motivate examining earnings composition, although establishing a credible group comparison and translating the measured responses into the model remain empirical tasks that I am currently working on. The goal of continued research would be to estimate the local spending generated through income feedback per dollar of wealth gain, and to establish what additional restrictions would be required to connect this estimate to the national missing-intercept problem.

2 A toy model: recovering what a difference leaves out

Consider two equally sized household types. Type 1 spends a fraction b_1 of each additional dollar of disposable income and type 2 spends b_2, where $0 < b_2 < b_1 < 1$. A programme gives $\tau > 0$ to each type-1 household and taxes each type-2 household by the same amount. The programme raises initial demand because it transfers resources to households that spend a larger fraction of them, even though the transfer itself leaves average income unchanged. The aim is to identify how much of the programme's aggregate consumption response comes directly from the transfers and how much comes from the additional earnings generated by the resulting spending.

Let c_i be type i's consumption response and Y the resulting income increase per household. Type i receives $a_i Y$ of this additional income, where the unknown shares satisfy $a_i \geq 0$ and $(a_1 + a_2)/2 = 1$. Prices and interest rates are fixed, firms meet demand, and consumption is the only expenditure component that changes. Household spending and market clearing then give

$$c_1 = b_1(\tau + a_1 Y), \qquad c_2 = b_2(-\tau + a_2 Y), \qquad Y = \frac{c_1 + c_2}{2}. \tag{1}$$

The unknown income shares matter: directing more of the earnings gain to type 1 creates more spending in subsequent rounds. Knowing the transfers and MPCs therefore does not by itself determine the aggregate response.

Suppose a research design identifies the programme's causal spending difference, $c_1 - c_2$, but cannot identify a movement common to both types. Dividing each consumption response by its MPC recovers the disposable income change that supports it. Averaging across households gives

$$\frac{1}{2}\left(\frac{c_1}{b_1} + \frac{c_2}{b_2}\right) = \frac{1}{2}\left[\tau + a_1 Y - \tau + a_2 Y\right] = Y = \frac{c_1 + c_2}{2}. \tag{2}$$

The transfers cancel because their financing is balanced, and the income shares sum to all additional income. The left-hand side measures the income needed to support spending; the right-hand side measures the income created by that spending. This equality supplies the second restriction needed to recover the two responses from their observed difference.

Substituting $c_1 = Y + (c_1 - c_2)/2$ and $c_2 = Y - (c_1 - c_2)/2$ into (2) yields

$$Y = \frac{b_1 - b_2}{2(b_1 + b_2 - 2b_1 b_2)}(c_1 - c_2). \tag{3}$$

The denominator is positive because $b_1 + b_2 - 2b_1 b_2 = b_1(1 - b_2) + b_2(1 - b_1) > 0$. Adding the same amount to both consumption responses would preserve their observed difference, but would change the income needed to finance them by more than the income generated by their extra purchases, since both MPCs are below one. The model selects the common response consistent with both household spending and production. Subtracting initial demand, $(b_1 - b_2)\tau/2$, then gives spending due to income feedback.

$$\underbrace{\frac{b_1 a_1 + b_2 a_2}{2} Y}_{\text{income feedback}} = \underbrace{Y}_{\text{recovered total}} - \underbrace{\frac{(b_1 - b_2)\tau}{2}}_{\text{known direct response}}. \tag{4}$$

The example makes precise what the model contributes, but also what the data must identify. The MPCs must apply to both transfers and earnings, and $c_1 - c_2$ must include the programme's differential equilibrium effects. A household experiment measuring only the response to an individual's own transfer does not automatically identify that difference. The regional application uses payroll and labour responses to obtain information about the earnings generated by the initial spending increase.

3 A simple regional model of income feedback

The regional model I present below is a simple adaptation of Chodorow-Reich et al. (2021). I retain stockholders, hand-to-mouth households and local spending, while simplifying production and wage adjustment. In their paper, they separate direct wealth spending from income feedback using an externally calibrated local multiplier. I propose identifying the same decomposition from differences in regional wage and hours responses, using the model's restrictions on who receives additional earnings and how much they spend. The model describes a relative wealth experiment across small regions, holding national prices, interest rates and future earnings fixed.

3.1 Who receives the additional earnings?

I first describe how additional earnings are distributed across households and how this determines the spending they generate. A share $\theta \in (0,1)$ of each region's households lives hand to mouth and spends all current earnings. The remaining households hold stocks and spend a fraction $\rho \in (0,1)$ of additional wealth or current earnings. I assume the same spending propensity out of both sources; Appendix A.1 gives a household problem that delivers this rule. Stocks are claims outside the region, so additional local earnings do not generate a further dividend payment.

Both types initially work $\bar{L}$ hours per household at wage $\bar{W}$. Following Chodorow-Reich, Nenov and Simsek (2021),[1] I assume that stockholders keep their hours fixed, while hand-to-mouth households accommodate the extra hours assigned by firms. Additional hours therefore generate income for households that spend all of it. Higher wages instead raise both types' earnings, including those of stockholders who save part of the gain. How payroll growth divides between hours and wages consequently matters for further spending.

Normalise the mass of households to one and let E denote total nominal household consumption expenditure in the region. Let x denote the wealth gain as a fraction of baseline payroll $\bar{W}\bar{L}$. The proportional changes in total hours and the wage are $\ell = dL/\bar{L}$ and $w = dW/\bar{W}$, respectively.

Hand-to-mouth households receive all additional hours income and a share θ of the income generated by higher wages. Their earnings gain, relative to baseline payroll, is therefore $\ell + \theta w$ to first order. Stockholders receive the remaining wage income, $(1-\theta)w$, alongside the initial wealth gain. Applying each type's spending propensity gives

$$\begin{aligned} \frac{dE}{\bar{W}\bar{L}} &= \underbrace{\ell + \theta w}_{\text{hand-to-mouth spending}} + \underbrace{\rho x + \rho(1-\theta)w}_{\text{stockholder spending}} \\ &= \underbrace{\rho x}_{\text{direct wealth spending}} + \underbrace{\ell + m_w w}_{\text{income feedback}}, \qquad m_w = \theta + (1-\theta)\rho. \end{aligned} \tag{5}$$

The coefficient on hours income is one because its recipients spend the whole gain. The coefficient $m_w \in (\rho, 1)$ is the average propensity to spend additional wage income, weighted by households' initial earnings. The object of interest is the income-feedback component, $\ell + m_w w$: additional spending financed by the local earnings response.

3.2 How does spending return as income?

I next describe how household spending generates local earnings, completing the feedback between income and consumption. Households spend a fixed share $\eta \in (0,1)$ on local nontradables and the remainder on tradables produced elsewhere. Competitive firms use labour alone, so nontradable payroll equals ηE. Outside buyers spend a fixed nominal amount on the region's tradable output, implying unit-elastic export demand and unchanged tradable payroll. At a balanced-trade benchmark, total payroll equals expenditure: $\bar{W}\bar{L} = \bar{E}$.

[1] See their Section IV.A and footnote 30, which explains how wage adjustment affects the distribution of additional earnings and hence the local multiplier.

Let n denote the proportional change in nontradable payroll. Since this payroll equals ηE, it grows at the same rate as household expenditure. Total payroll grows only through the nontradable sector, which initially accounts for a share η of total payroll. These conditions give, to first order,

$$n = \frac{dE}{\bar{E}} = \frac{dE}{\bar{W}\bar{L}}, \qquad \ell + w = \eta n. \tag{6}$$

Only the fraction η of additional spending therefore returns as local earnings; the remainder pays for imports. A wage rule $w = \kappa\ell$, with $\kappa \geq 0$, determines how payroll growth divides between wages and hours. Firms meet demand at the resulting wage, and hand-to-mouth households supply the assigned hours; I do not impose an additional labour-supply condition.

Together with the household spending rule in (5), these equations close the regional model. Successive rounds of spending and earnings generate a unique equilibrium, whose multiplier is derived in Appendix A.2.

4 What do the responses identify?

I now show when regional payroll and hours responses identify direct wealth spending and income feedback under the model's restrictions. Suppose a research design identifies the causal responses of nontradable payroll, total payroll and total hours to the normalised wealth shock: β^N, β^P and β^L. Outcomes are proportional changes, and all three responses concern the same horizon and population. Since payroll growth is wage growth plus hours growth to first order, $\beta^W = \beta^P - \beta^L$ measures the wage response. Combining the spending rule in (5) with market clearing in (6), and expressing responses per unit of the wealth shock, gives

$$\underbrace{\beta^N}_{\text{total spending per wealth dollar}} = \underbrace{\rho}_{\text{direct spending}} + \underbrace{\beta^L + m_w\beta^W}_{\text{income feedback}}. \tag{7}$$

The normalisation makes the three terms comparable: β^N measures additional spending per dollar of wealth, while β^L and β^W measure additional earnings from hours and wages per dollar of wealth. The target is $\beta^N - \rho$, the amount of spending financed by those earnings. It is a nominal expenditure response; recovering real consumption would also require the price response.

With fixed wages, all additional earnings come from extra hours and reach hand-to-mouth households, who spend the entire gain. In that case $\beta^W = 0$ and $\beta^P = \beta^L$, so feedback equals the payroll response and $\rho = \beta^N - \beta^P$. When wages also rise, some earnings reach stockholders and a single regional response leaves two unknowns, ρ and m_w, in one equation.

To separate them, I compare two groups of regions $g \in \{1, 2\}$ with common household parameters ρ, θ and expenditure share η, but different wage-adjustment parameters κ_g. Both groups therefore have the same spending propensities. Their earnings gains can differ in composition. After subtracting spending financed by hours, define $Z_g = \beta^N_g - \beta^L_g$. The two groups supply

$$Z_1 = \rho + m_w\beta^W_1, \qquad Z_2 = \rho + m_w\beta^W_2. \tag{8}$$

Taking the difference between groups removes the common direct wealth effect. The remaining spending difference is financed by additional wage income, so dividing it by the difference in wage-income responses identifies m_w. Subtracting this wage-financed spending from either group's Z_g then recovers ρ.

Proposition 1 (Identification from earnings composition)**.** *Under the regional model, identified causal responses in two groups with $\beta_1^W \neq \beta_2^W$ identify the direct MPC and income feedback:*

$$m_w = \frac{Z_2 - Z_1}{\beta_2^W - \beta_1^W}, \qquad \rho = Z_1 - m_w \beta_1^W, \qquad \textit{feedback in group } g = \beta_g^N - \rho. \tag{9}$$

Proof. Subtract the equations in (8) to recover m_w, then substitute into either equation to recover ρ. Equation (7) gives the remaining spending component. □

Different wage rules generate different wage responses in this model, as Appendix A.2 verifies. Recovery requires common household spending behaviour and the assumption that recipients spend all additional hours income. If MPCs or household shares differ freely across groups, earnings composition no longer isolates the unknown propensities. A small wage-response difference also makes the calculation imprecise. The recovered values must satisfy $0 < \rho < m_w < 1$, and the model requires $\beta_g^P = \eta \beta_g^N$; these implications provide checks on the proposed interpretation.

4.1 Sensitivity to spending out of hours income

I next examine how identification depends on the assumption that households spend all additional hours income. Suppose the households receiving those hours instead spend a common fraction m of additional earnings, where $\rho < m \leq 1$. They are no longer strictly hand to mouth when $m < 1$, but the allocation of hours and wages is unchanged. The spending equation becomes

$$\beta_g^N = \rho + m\beta_g^L + m_w \beta_g^W, \qquad m_w = \theta m + (1 - \theta)\rho. \tag{10}$$

For a given m, subtracting $m\beta_g^L$ from each group's spending response again leaves two equations in ρ and m_w, which can be solved as in (9). I can therefore examine how the inferred direct MPC and feedback change across externally justified values of this propensity, retaining only values consistent with $0 < \rho < m_w < m$.

If m is also unknown, different spending propensities can reproduce the same observed responses. With a common local spending share η and an unknown household share θ, market clearing links hours and wages so that additional wage-adjustment groups cannot resolve this ambiguity. Appendix A.3 shows the trade-off: among admissible values consistent with the same responses, a lower m implies a larger direct wealth effect and less income feedback. The empirical exercise must therefore either justify the hours-income propensity or report the decomposition's sensitivity to it.

5 An empirical illustration and the next comparison

Chodorow-Reich et al. (2021) estimate local labour-market responses to stock-wealth gains and use external evidence on the local multiplier and labour share to recover a direct wealth MPC. Using

their replication data, I reproduce the payroll and employment responses and examine what these outcomes can tell us about the income feedback described above.

5.1 What is estimated?

For county a, shock quarter t and horizon h, let $q^k_{a,t+h}$ be the log change in outcome k from $t-1$ to $t+h$. I estimate

$$q^k_{a,t+h} = \alpha^k_{a,h} + \delta^k_{\text{state}(a),t,h} + \beta^k_h x_{at} + C'_{at,h}\gamma^k_h + u^k_{a,t+h}. \tag{11}$$

The wealth shifter x_{at} is lagged county stock wealth multiplied by the portfolio-adjusted stock return, divided by annual IRS wage income. County effects absorb persistent local differences and state-by-quarter effects absorb movements shared within a state. I retain the authors' controls and population weights, pooling shock dates from 1992Q3 to 2015Q2. Interpreting the estimates as local wealth responses requires conditional wealth exposure not to predict responses to other shocks or differential responses to the national consequences of the stock-market shock itself (Manuel, 2026).

Table 1 reports the responses at the two-year horizon. These measure changes from $t-1$ to $t+7$, rather than cumulative earnings over those quarters.

Table 1: Full-sample responses to stock-wealth gains

Outcome	Estimated response
Nontradable payroll	3.24 (1.01)
Total payroll	2.19 (0.63)
Total employment	0.78 (0.36)
Average earnings per worker	1.41 (0.49)

Notes: Responses in basis points to a stock-wealth gain equal to 1% of annual county IRS wage income, measured from $t-1$ to $t+7$. Standard errors clustered by county and quarter appear in parentheses. All outcomes use the same sample of 2,896 counties and 263,210 county-quarter observations, controls and population weights.

5.2 What does the evidence contribute?

The positive nontradable and total payroll responses show that stock-wealth gains generate additional local earnings. Figure 1 shows how these responses develop over time: payroll and employment increase over the quarters following the wealth gain, and the earnings increase persists beyond the initial quarter. Earnings per worker also rise, so employment growth accounts for only part of the payroll response.

Because the outcomes use the same sample and specification, the payroll response decomposes exactly into employment and average earnings per worker: $\log P = \log J + \log(P/J)$, where P is total payroll and J is employment. At the two-year horizon, 0.78 of the 2.19 basis points comes from employment and the remaining 1.41 from earnings per worker. The latter includes hours per worker, wages and workforce composition, so it cannot yet be interpreted as the model's wage response.

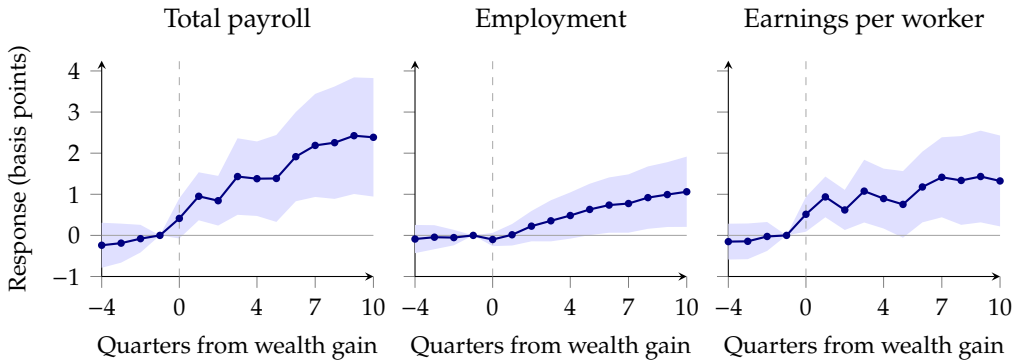


Figure 1: Payroll, employment and earnings per worker over time

Notes: Responses in basis points to a wealth gain equal to 1% of annual county IRS wage income. Outcomes are log changes relative to $t-1$, whose response is zero by construction. Shading denotes pointwise 95% confidence intervals, with county-and-quarter clustered standard errors. Each horizon uses the same sample across outcomes; observations vary slightly across horizons because of data availability. Earnings per worker includes hours, wage and workforce-composition changes.

I have not yet worked through the comparison between groups fully. Building on the pooled estimates above, I outline the regressions I would run to apply the model's identification argument. For each outcome k, comprising nontradable payroll, total payroll and employment, I would estimate

$$q^k_{a,t+h} = \alpha^k_{a,h} + \delta^k_{\text{state}(a),t,h} + \beta^k_{1,h} x_{at} + \lambda^k_h D_a x_{at} + C'_{at,h}\gamma^k_h + u^k_{a,t+h}. \tag{12}$$

The indicator D_a equals one for counties in the second group, so the estimated responses are $\beta^k_{1,h}$ and $\beta^k_{1,h} + \lambda^k_h$. I would use the same observations across outcomes and retain the existing controls, population weights and standard errors clustered by county and quarter.

These estimates provide the group responses needed for the decomposition in Section 4. In the simplified model, subtracting spending financed by additional hours leaves direct wealth spending and spending out of higher wages:

$$Z_g = \beta^N_g - \beta^L_g = \rho + m_w \beta^W_g.$$

Taking the difference between groups removes the common direct wealth effect ρ. Dividing the remaining spending difference by the difference in wage-income responses therefore identifies m_w. Subtracting $m_w \beta^W_g$ from either group's Z_g then recovers ρ, and total income feedback equals $\beta^N_g - \rho$. This recovery assumes common household spending propensities across groups. Applying it to the data requires the hours, workforce-composition and labour-share adjustments discussed in Appendix A.5.

6 Preliminary conclusion

I conclude by summarising what this work in progress establishes and outlining the next steps. The toy model shows how household spending rules and market clearing can identify income feedback: the transfers and known MPCs determine direct spending, while the observed consumption difference and the equilibrium restriction recover total consumption. The regional model considers a setting in which the direct wealth MPC is unknown. Under its assumptions, differences in wage

and hours responses across regions with common spending behaviour provide the additional information needed to separate direct spending from income feedback.

The next step is to establish whether the available data support such a comparison, particularly whether regions with different wage adjustment have sufficiently comparable household spending behaviour. If the comparison supports the model's restrictions, I would estimate direct spending and income feedback, accounting for the measurement issues discussed above and reporting uncertainty and sensitivity to the assumptions needed for recovery. This would allow me to assess whether regional responses can identify the spending financed by additional earnings without supplying an external local multiplier.

References

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A Supporting derivations and measurement

A.1 A household foundation for the spending rule

A stockholder chooses current and future nominal expenditure, taking prices P_0, P_1, the gross interest rate R and lifetime resources V^s as given:

$$\max \ \rho \log(E_0^s/P_0) + (1-\rho)\log(E_1^s/P_1) \quad \text{subject to} \quad E_0^s + E_1^s/R = V^s. \tag{A.1}$$

The first-order condition gives $E_1^s/R = (1-\rho)E_0^s/\rho$, hence $E_0^s = \rho V^s$: an extra dollar of wealth or current earnings raises spending by ρ when future earnings are fixed. Hand-to-mouth households spend current earnings and receive all additional hours: $L = \theta L^h + (1-\theta)\bar{L}$ implies $dL = \theta dL^h$. Differentiating earnings gives the allocation in (5). The rule $W = \bar{W}(L/\bar{L})^\kappa$ implements wage adjustment, with firms assigning hours and households accommodating them.

A.2 Equilibrium and the identifying wage difference

In the household toy, substituting the spending rules into market clearing gives $Y = (b_1 - b_2)\tau/(2 - b_1a_1 - b_2a_2)$. The denominator is positive because $a_1 + a_2 = 2$ and both MPCs are below one, so the equilibrium used in Section 2 exists uniquely for every admissible income allocation.

In the regional model, payroll accounting and the wage rule give $\ell = \eta n/(1+\kappa)$ and $w = \kappa\eta n/(1+\kappa)$. Substitution into spending yields

$$n = \rho \mathcal{M} x, \qquad \mathcal{M} = \left[1 - \eta\frac{1+\kappa m_w}{1+\kappa}\right]^{-1}. \tag{A.2}$$

The feedback coefficient is positive and at most $\eta < 1$, so the first-order equilibrium is unique. Greater wage adjustment reduces amplification by directing more income to stockholders who save part of it. In this model the target $\beta_g^N - \rho$ equals $\rho(\mathcal{M}_g - 1)$.

The wage response and its derivative are

$$\beta_g^W = \frac{\eta\rho\kappa_g}{(1-\eta) + \kappa_g(1-\eta m_w)}, \qquad \frac{\partial \beta_g^W}{\partial \kappa_g} = \frac{\eta\rho(1-\eta)}{[(1-\eta) + \kappa_g(1-\eta m_w)]^2} > 0. \tag{A.3}$$

Thus different κ_g, with common household parameters and spending shares, generate the wage-response contrast required for identification.

A.3 What changes when hours recipients save?

Let the recipients of additional hours spend fraction m of current earnings, with $\rho < m \leq 1$. The multiplier becomes $[1 - \eta(m + \kappa m_w)/(1+\kappa)]^{-1}$, which remains positive and finite because $0 < m_w < m \leq 1$ and $\eta < 1$. Using $\beta_g^L = \eta\beta_g^N - \beta_g^W$ in (10) gives

$$\beta_g^N = A + B\beta_g^W, \qquad A = \frac{\rho}{1-m\eta}, \qquad B = \frac{m_w - m}{1-m\eta}. \tag{A.4}$$

Distinct wage responses identify the intercept A and slope B; market clearing identifies $\eta = \beta_g^P/\beta_g^N$. For any candidate m, the same responses are consistent with

$$\rho(m) = (1 - m\eta)A, \qquad m_w(m) = m + (1 - m\eta)B, \qquad \theta(m) = \frac{m_w(m) - \rho(m)}{m - \rho(m)}, \tag{A.5}$$

provided $0 < \rho(m) < m_w(m) < m \leq 1$. These inequalities ensure an admissible household share and a unique equilibrium reproducing the responses with unchanged wage rules. Around an interior solution, several m satisfy them; further groups with the same η trace the same line. Since $A > 0$, lowering m raises $\rho(m)$ and lowers feedback $\beta_g^N - \rho(m)$. Differences in feedback still equal $\beta_2^N - \beta_1^N$ under a common direct MPC.

A.4 Estimating the group comparison

For two groups, the regression in (11) replaces $\beta_h^k x_{at}$ with $\sum_{g=1}^{2} \beta_{g,h}^k \mathbf{1}\{g(a) = g\} x_{at}$, allowing the wealth response to differ across them. A predetermined grouping alone does not justify common MPCs: characteristics affecting wage adjustment may also reflect household resources and stock ownership. I would assess these differences alongside the earnings responses, with controls allowing for differential exposure to national developments.

Before calculating a decomposition, I would estimate the group responses jointly, check the precision of the wage-response contrast and assess the model's common spending-share restriction. Employment must be translated into hours, and workforce composition accounted for before residual payroll growth can measure wage growth. Allowing capital also requires a labour share to translate nontradable payroll into spending. Appendix A.5 explains these adjustments and the additional timing assumptions needed to connect a two-year response to the model.

A.5 Hours, capital and empirical normalisation

With an independently justified hours-to-employment response ratio χ_g, total hours and wages would be constructed as

$$\beta_g^L = \chi_g \beta_g^{\text{emp}}, \qquad \beta_g^W = \beta_g^P - \chi_g \beta_g^{\text{emp}}. \tag{A.6}$$

The original paper uses $\chi = 1.5$. Applying it across groups assumes comparable hours-per-worker adjustment; composition changes must also be accounted for before interpreting the residual as a common wage. Table 1 imposes no value of χ.

With constant sectoral labour shares s_N, s_T, nontradable payroll is $s_N \eta E$. At the balanced-trade benchmark, total payroll is $\bar{s}\bar{E}$, where $\bar{s} = \eta s_N + (1 - \eta) s_T$. Nontradable payroll still changes proportionally with expenditure, so

$$\frac{\beta_g^N}{\bar{s}} = \rho + \beta_g^L + m_w \beta_g^W, \qquad \text{feedback in group } g = \frac{\beta_g^N}{\bar{s}} - \rho. \tag{A.7}$$

The two-equation argument survives conditional on the labour share and the household-income assumptions. This adjustment does not implement the original paper's full national model.

The regressions use annual IRS wage income and quarterly payroll growth. A structural interpretation requires consistent flow units and income populations, and measured nontradable industries

that represent the model's sector. The $h = 7$ estimate is an endpoint response, not cumulative spending. Mapping it into the model's single period additionally requires assumptions about intervening saving and responses to expected future income; matching units alone does not justify the static decomposition.

Figure 1 retains the horizon-specific industry control and uses common observations across outcomes at each horizon. Regressing log payroll minus log employment directly incorporates their covariance in the earnings-per-worker standard error. Pointwise intervals use the county-and-quarter clustered standard errors and corresponding t critical value; the $h = -1$ response is zero by construction.